Mike Hall

Senior / Lead Software Engineer, Ruby on Rails

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Professional Summary

Senior software engineer focused on Ruby on Rails infrastructure, PostgreSQL, legacy modernization, and production reliability. Delivers business value through complex state machine refactoring, database optimization, asynchronous processing, third-party integrations, and disciplined testing.

Core Skills

Ruby on Rails (Full-Lifecycle 2.x through 8.x), PostgreSQL, pgvector & MySQL Query Optimization & Indexing, Asynchronous Processing (Solid Queue, Sidekiq, Resque, Redis), State Machine Refactoring & Data Integrity, Third-Party API & Webhook Integrations, Test-Driven Development (RSpec, Capybara, FactoryBot), Docker, Linux, Git & Zero-Downtime Deployment, Fast Ramp-Up & Autonomous IC Execution

Experience

Staff Software Engineer

just3ws (Independent) | Chicago, IL | February 2026 - Present

Architected and operating Rails 8, Solid Queue, and PostgreSQL 18 (pgvector) applications featuring hybrid vector/full-text search, RubyLLM local AI synthesis, YouTube API v3 resumable publishing, and Ahoy analytics.

- **Autonomous Media Pipeline & Local AI RAG Microservice:** Architected a Rails 8, Solid Queue, and PostgreSQL 18 (pgvector) media ingestion and publishing platform with on-device Metal GPU Whisper transcription, hybrid lexical and vector semantic search via Reciprocal Rank Fusion (RRF), local LLaMA 3.1 RAG synthesis via RubyLLM, OpenTelemetry distributed tracing, Ahoy analytics, and PgHero query tuning across 270+ processed video masters.
- **Forensic Audio DSP Restoration & Sovereign Verification (dehum):** Engineered a deterministic 5-stage FFmpeg audio restoration and clock-drift synchronization pipeline (atempo, adeclick, highpass, afftdn, and 8-harmonic 60 Hz notch cascade) eliminating electromagnetic interference and corona discharge crackle recorded under 345,000-volt transmission lines while preserving lossless video streams.
- **Technical Conversation Archive & Knowledge Graph:** Built and curated a contract-validated oral history platform preserving 218 technical interviews with Software Craftsmanship, Ruby, and JVM practitioners, enforcing build-time schema verification, speaker diarization, and semantic knowledge graph cross-linking across 500+ indexed pages.
- **Deterministic State Engine & Verification Platform (Phalanx Duel):** Engineered a server-authoritative real-time tactical game platform in TypeScript, Node.js, PostgreSQL, and WebSockets featuring a pure deterministic rules engine, append-only replayable action ledger, OpenTelemetry trace propagation, and property-based CI verification gates across the complete state lifecycle.
- **Multi-Source Ingestion & Prompt-Injection Defense (WWWorkRemote):** Designed a Rails 8 and pgvector semantic matching platform with a Manifest V3 Chrome extension across 16 ATS adapters and a 4-stage prompt-injection guardrails pipeline screening untrusted scraped text before local LLM evaluation.

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Architected and delivered an instant loan disbursement pipeline (funding to debit cards in minutes) and refactored multi-step Rails workflow state machines, eliminating lateral state corruptions across high-volume customer prequalification pipelines.

- **Originations IC Delivery & Instant Disbursement:** Led the Originations Verification squad through consecutive Exceeds Expectations ratings, architecting and shipping an instant loan disbursement pipeline (funding to debit cards in minutes). Appointed Software Architect for the Acquisition Lane, later converting to Associate Director, Staff Engineer.
- **Acquisition Workflow Enablement & Architecture Discovery:** Founded the Acquisition Lane enablement team; mapped the loan acquisition and origination pipeline end-to-end from visit to booked loan (desktop browser to mainframe and back again at each step) and the Affiliate API at the same depth, identifying seven distinct lending funnels with a tracking initiative; architected an automated 5-phase PII remediation deletion engine across 30+ tables.
- **Enterprise Resilience & DynamoDB Session Remediation:** Diagnosed and eliminated a critical session-cookie overflow defect that caused 4% silent traffic loss of digital loan applications during e-signing; architected zero-downtime blue/green migration to DynamoDB session store with zero incidents.
- **Enterprise Trace & Operational Alignment:** Architected distributed telemetry across Rails distributed monoliths, MuleSoft APIs, and IBM mainframe backends; partnered directly with Cybersecurity, EMC, SRE, and Incident Command to align monitoring with end-to-end distributed tracing.

- **Community Enablement & SRE Handoff:** Founded weekly Geekfest@OMF and scaled the enterprise OpenTelemetry Working Group to 40+ cross-lane engineers; mentored and partnered with SRE leads to transition ongoing operational facilitation, establishing a durable community of practice.
- **Applied AI & Innovation Leadership:** Placed in two corporate hackathons building conversational agents with Rasa and a schema inference prototype; pioneered local LLM orchestration and privacy-conscious AI developer tooling at Geekfest@OMF.

Senior Backend Developer

SK Holdings, Inc. | Chicago, IL | January 2019 - December 2020

Sequenced zero-downtime Ruby and Rails version upgrades, optimized MySQL full-text search indices, and built batch Sidekiq/SendGrid delivery pipelines.

- **Search Infrastructure Modernization:** Eliminated external cluster failure modes and reduced operating footprint by migrating legacy Sphinx to optimized MySQL full-text search and relational indices under live production traffic.
- **Zero-Downtime Rails Upgrades:** Led Ruby and Rails framework upgrades across core applications, sequencing database migrations to ensure zero downtime during platform modernization.

Senior Software Developer

ActiveCampaign | Chicago, IL | September 2018 - December 2018

Introduced testable query patterns and caching into a tightly coupled PHP/MySQL CRM backend and extended the Ember.js frontend test suite across contact management workflows.

- **Legacy Query Modernization:** Replaced ad-hoc global data access with structured, cacheable query patterns in the PHP backend, enabling unit testability and sub-second query execution.
- **Frontend Verification Gates:** Hardened Ember.js automated test suites across core CRM contact workflows, creating safety gates that enabled non-breaking legacy PHP refactoring.

Senior Software Developer

BenchPrep | Chicago, IL | March 2017 - February 2018

Optimized complex PostgreSQL exam scoring calculations, implemented JWT cryptographic response verification, and built secure embedded client APIs in Rails.

- **Assessment Query Optimization:** Optimized complex PostgreSQL calculations across high-stakes exam scoring engines, eliminating calculation drift and reducing query latency.
- **JWT Response Verification:** Implemented JWT-based cryptographic payload signing and once-only processing, securing exam submission pipelines against client-side tampering.
- **Secure Embedded Client APIs:** Designed secure cross-origin API boundaries for embedded enterprise assessment clients, establishing explicit authentication and trust contracts.

Senior Software Developer

ReachLocal | Remote | March 2015 - November 2016

Led phased Strangler-Fig legacy migration and built an internal Rails API Gateway with token authentication.

- **Strangler-Fig Modernization:** Architected incremental modernization roadmap, replacing a high-risk full rewrite with a phased Strangler-Fig migration under production load.
- **Internal API Gateway:** Built internal API Gateway with standardized JSON contracts and token authentication, safely isolating legacy backend services.
- **Automated CI/CD Verification:** Introduced automated static analysis and security scanning into deployment pipelines, preventing regression and vulnerability drift.

Open-Source Transition Lead

Coderwall | Remote | January 2014 - December 2014

Led the open-source transition of a YC-backed Rails developer network, closing SQLi/XSS/DoS vulnerabilities and migrating from MongoDB and Resque to PostgreSQL and Sidekiq.

- **Open-Source Transition & Security Hardening:** Led open-source transition for a YC-backed developer network (856 GitHub stars, 304 forks), closing critical SQL injection, XSS, and Symbol DoS vulnerabilities before public release.
- **Proprietary Service Decoupling:** Extracted proprietary billing engines and scoring algorithms into isolated backend services with clean API stubs, safeguarding business logic.
- **Database Migration & Sidekiq Modernization:** Executed two-phase database migration from MongoDB to PostgreSQL and converted legacy Resque jobs to Sidekiq, cutting operational costs.

Software Engineer

Obtiva | Chicago, IL | August 2009 - July 2011

Built production Ruby on Rails applications, B2B catalog services, and high-growth deal processing pipelines for Sears, Leapfrog Online, and Groupon.

- **Self-Service Analytics Platforms:** Built reporting platforms for Leapfrog Online, enabling business stakeholders to query and export data warehouse analytics autonomously.
- **Enterprise B2B Catalog Services:** Developed commercial B2B sales and product catalog integration services for Sears, opening new enterprise revenue streams.
- **Hypergrowth Transaction Pipelines:** Engineered deal-processing services and fraud analysis pipelines at Groupon, stabilizing backend transaction infrastructure during extreme user growth.

Earlier Career Experience

- **Principal Consultant** | **Tandem** | Chicago, IL | August 2018 - August 2018: Conducted an on-site architectural and operational assessment for an at-risk federal software program (DoD MEPS), delivering viability analysis and risk mitigation recommendations directly to executive leadership.
- **CTO** | **KloboMedia** | Remote | March 2014 - September 2016: Owned end-to-end platform architecture, leading high-throughput data integration for an early-stage social media startup. The product, TheSocReport, ended when the upstream data taps closed: Twitter's firehose shutdown and the closing of Facebook's and Instagram's user-intelligence APIs removed the inputs the analytics depended on.
- **Senior Software Developer** | **Upcity** | Remote | October 2013 - February 2014: Owned payment integrations and local development infrastructure, leading system modernization for an SOA-based marketing platform.
- **Senior Software Developer** | **Viewpoints** | Chicago, IL | May 2013 - October 2013: Owned core business system development, leading AWS deployment modernization and CI implementation.
- **Software Engineer & Technical Onboarding Lead (Fraud & Taxonomy Systems)** | **Groupon** | Chicago, IL | July 2011 - May 2013: Owned core backend systems for fraud and taxonomy, leading global engineering enablement during hyper-growth.
- **Principal Developer** | **Business Decisions, Inc.** | Remote | October 2008 - August 2009: Owned all technical strategy and system development, leading lifecycle modernization for multi-firm consulting teams.
- **Assistant Vice President, Application Developer** | **JPMorgan Chase & Co.** | Chicago, IL | April 2008 - October 2008: Owned cross-system integration stability, leading defect resolution in regulated financial workflows.
- **Senior .NET Developer** | **Brightstar Corporation** | Libertyville, IL | March 2007 - April 2008: Owned enterprise supply-chain integration, leading cross-system data unification for a global mobile device distributor.
- **Senior .NET Developer** | **TicketsNow** | Crystal Lake, IL | November 2005 - March 2007: Owned real-time inventory systems, leading transactional integrity and iterative delivery for revenue-critical operations.

Technical Leadership

- Established enterprise-wide OpenTelemetry standards and transitioned ownership after adoption stabilized across teams
- Realigned Acquisition engineering responsibilities around business ownership, then founded and technically led an enablement team to sustain platform risk and remediation work
- Built Communities of Practice to prevent architectural drift and align cross-team decisions
- Built and ran global engineering onboarding and training at Groupon during hypergrowth, improving ramp time and retention
- Founded Software Craftsmanship McHenry County (2009) with Ryan Gerry and Jim Suchy, building a community that still meets with 820+ members
- Founded WHOIS Tech Community (UGtastic), interviewing industry leaders and shaping software craftsmanship discourse